ISHIKA KANAKATH

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PROFESSIONAL SUMMARY

PhD student in Electrical and Computer Engineering at Arizona State University in Prof. Krishnendu Chakrabarty's group, working on security assurance for heterogeneous chiplet packages and adversarial security of ML systems. Intel ISFP Fellow. Diverse background in RTL design and verification, RTL-to-GDS physical design, neuroengineering, control theory, and optimization, with two master's degrees and industry internships at MACOM and Cadence.

EDUCATION

Arizona State University — **2026–Present**
PhD, Electrical and Computer Engineering | Advisor: Krishnendu Chakrabarty | Tempe, AZ

University of Washington — **2026**
MS, Electrical Engineering | 4.0 GPA | Seattle, WA

Case Western Reserve University — **2025**
MS, Computer Engineering | 3.9 GPA | Cleveland, OH

Case Western Reserve University — **2025**
BS, Computer Engineering | Cleveland, OH

Relevant coursework: Embedded Systems; Digital Circuits and Systems; VLSI Design; High-Performance Data and Computing; FPGA Design; Quantum Computing; Haptics and Robotics; Convex Optimization; First-Order Methods in Optimization; Control Systems Theory; Agentic Control Systems Theory; Linear Multivariable Control; Robust Control; Neural Engineering; Advanced Neurotechnology; Statistical and Adaptive Signal Processing

FELLOWSHIPS & FUNDING

- Intel Scholarship and Fellowship Program (ISFP) Fellowship, managed by SRC; sole PhD student on an Intel-funded SRC package-assurance project starting January 2027. Also supported under the group's NSF award on privacy-preserving AI for physiological data.

RESEARCH EXPERIENCE

Graduate Research Assistant — Arizona State University, Chakrabarty Group — **2026–Present**

- Package assurance (ATLAS): security assurance for 2.5D/3D heterogeneous chiplet packages, covering threat modeling, hardware trust, runtime monitoring, and security coverage, under an SRC/Intel-funded project. Support the companion PATH effort on package test, diagnosis, repair, and lifecycle health.
- First-author paper (target: ETS 2027) on the integrity of die-to-die link test and repair: how supply droop coupled across a shared PDN can corrupt the UCIe lane-repair verdict during post-assembly test, causing spare-lane misallocation or a forced width degrade. Built and VCS-verified a spec-faithful UCIe Advanced Package spare-lane remap block and integrating it into the Berkeley UCIe PHY RTL, with a Voltus and PrimeTime flow tying droop to repair outcome and an on-chip droop sensor as the defense.
- Adversarial security of ECG-language models (with Prof. Farshad Firouzi): adaptive, physiologically constrained attacks on the self-verification stage of an agentic clinical ECG-LLM pipeline (signal perturbation, retrieval poisoning, context injection), and waveform-level anonymizers as an input-side defense. Target: BSN/CHASE.

Graduate Research Assistant — University of Washington — **2025–2026**

- Constrained sensor selection under adaptive drift: which recording channels to stream under telemetry, power, and bandwidth budgets when the signal source and the decoder adapt simultaneously. Developed a two-budget framework with regret bounds drawing on submodular optimization, online learning with switching costs, and performative prediction. Theory paper in preparation for arXiv.
- Showed the SmoothBatch decoder update used across neural interfaces is damped Newton on the static decoder problem, and that a two-process model of human learning makes the coupled human-decoder system second-order. Submission to L4DC 2027 with V. Charisopoulos and P. Patel.

Research Intern, NASA-Funded WAAM Project — Case Western Reserve University — **2023–2025**

- Designed a semi-closed-loop adaptive control system for wire arc additive manufacturing (laser scanner and motion controller integration, discrete-time plant modeling, P/PI design, z-plane stability analysis); 0.043 mm RMS precision, projected 35-fold reduction in cumulative height error versus open loop.

INDUSTRY EXPERIENCE

MACOM Technology Solutions — Digital Circuit and DSP Intern — **Lowell, MA | 2025**

- Developed and verified Verilog and SystemVerilog RTL modules for digital signal processing pipelines targeting ASIC integration, including DC offset removal, convergent rounding, and digital amplification stages.
- Performed functional simulation and timing analysis of DSP blocks and worked with design teams on synthesis and implementation for high-throughput, low-latency signal processing architectures.

Cadence Design Systems — Application Engineering Intern — **San Jose, CA | 2023**

- Carried a full chip design from RTL through post-route implementation, migrating it from GPDK45 to the open-source SKY130 process. Ran Genus for synthesis and Innovus for floorplanning, placement, clock tree synthesis, routing, and static timing analysis, closing timing and design-rule checks.

Experian — Software Engineering Intern and Technical Contractor — **Costa Mesa, CA | 2022 and 2025**

- Designed hardware-software interfaces and built production applications using generative AI, AWS (S3, EC2), data analytics, and microservice architectures in an agile team. Returned in 2025 as a technical contractor.

PUBLICATIONS AND MANUSCRIPTS

- I. Kanakath. Co-Adaptive Learning and Domain Adaptation Bridge the Translational Gap in Neuroprosthetic Control: A Review. Frontiers in Neuroscience (Neuroprosthetics), 2026 (accepted). doi:10.3389/fnins.2026.1974861

- Cross-die droop attacks on chiplet lane repair. Target: ETS 2027.
- Survey of security and trust in heterogeneous chiplet integration (with A. Hati, K. Chakrabarty). Target: ACM JETC.
- Budgeted selection under co-adaptive drift: theory paper. Target: arXiv.
- SmoothBatch as damped Newton in co-adaptive neural interfaces (with V. Charisopoulos, P. Patel). Target: L4DC 2027.

TEACHING & LEADERSHIP

- Co-Instructor and Teaching Assistant, EE 543 Models of Robot Manipulation, University of Washington, 2026.
- Teaching Assistant, ECE 474 Embedded Systems and Design and EE 271 Digital Circuits and Systems, University of Washington, 2024–2026.
- President, HackCWRU. First female president; raised $10,000 for Cleveland's largest student-run hackathon.

HONORS

Intel ISFP Fellowship, SRC (2026) · SOURCE STEM Scholarship (2024) · Robert L. Shurter Prize (2024) · Jerry P. Lind Prize (2024–2025) · ISAAC Scholarship (2024–2025) · Dean's Honors List, five semesters · University Merit Scholarship (2021–2025) · Semifinalist, UW Hollomon Health Innovation Challenge

TECHNICAL SKILLS

Hardware Design: SystemVerilog, Verilog, VHDL, TCL; RTL design and verification; synthesis; place and route; static timing analysis; Genus, Innovus, Xcelium, VCS, ModelSim, Synplicity, KiCad

Hardware Security: Threat modeling for chiplet and heterogeneous packages; secure boot and root-of-trust attestation; power-delivery fault and cross-die droop attacks; UCIe link training and lane repair; hardware trojan taxonomy (Trust-Hub); on-chip telemetry and silicon lifecycle monitoring; side-channel and supply-chain threat models

ML Security & Privacy: Adversarial attacks on LLM and RAG pipelines; physiologically constrained perturbations; federated learning inversion and poisoning; anonymization evaluation

Controls & Human-Machine Systems: Control theory (linear multivariable, robust, adaptive, and agentic control); dynamical systems; human-machine interaction; neural interfaces and brain-computer interfaces; co-adaptive learning

Programming & Methods: Python (NumPy, SciPy, PyTorch), MATLAB, C, CUDA, Java, Git, Linux, AWS, LaTeX, ROS; convex and submodular optimization; control and dynamical systems; state estimation; statistical signal processing; information theory